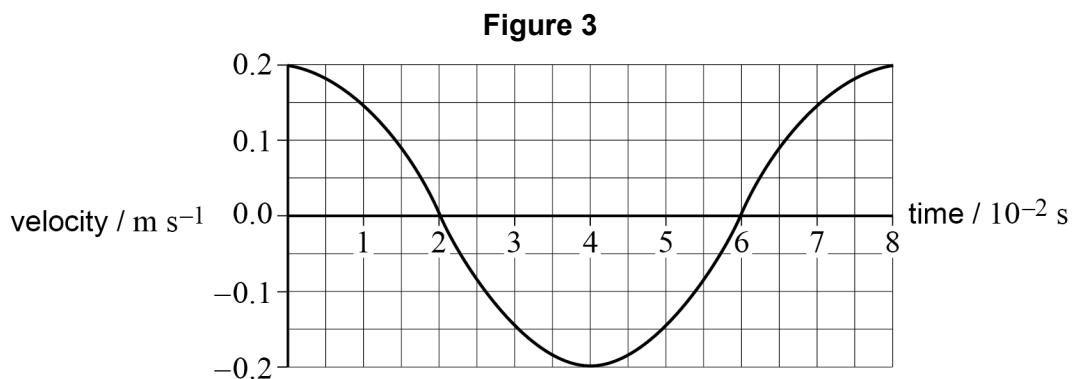


0 5 **Figure 3** and **Figure 4** are identical velocity–time graphs for an object oscillating about a fixed point.

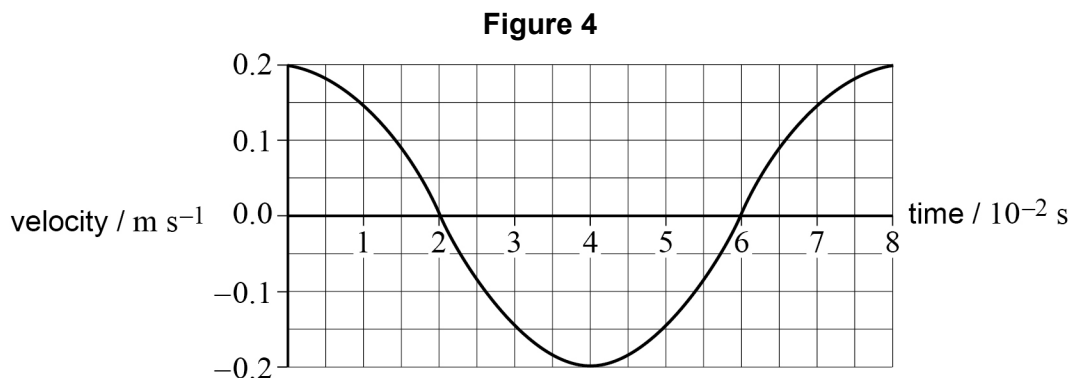
0 5 . 1 Determine using **Figure 3** the maximum acceleration of the oscillating object. **[3 marks]**



maximum acceleration = _____ m s^{-2}

0 5 . 2 Determine using **Figure 4** the maximum displacement of the object from the fixed point.

[3 marks]



maximum displacement = _____ m



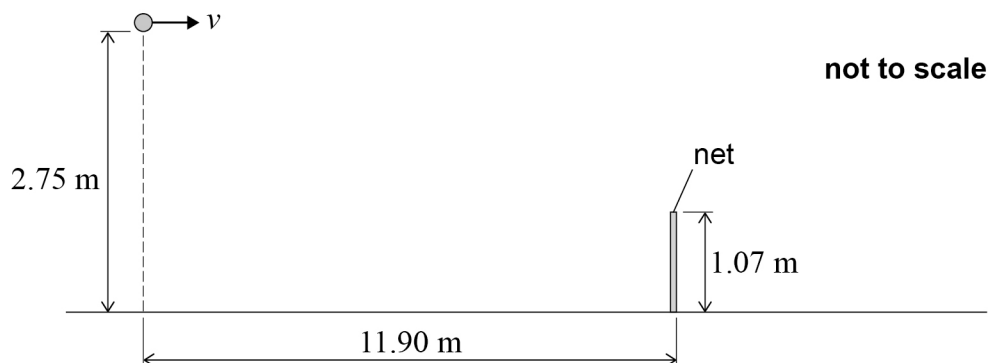
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0 6

A tennis ball is served horizontally as shown in **Figure 5**. The bottom of the ball is 2.75 m above the ground and a horizontal distance of 11.9 m from the net. The top of the net is 1.07 m above the ground.

Assume air resistance is negligible in this question.

Figure 5



0 6

1

Show that the time taken for the bottom of the ball to fall to 1.07 m above the ground is approximately 0.6 s

[2 marks]

0 6

2

Calculate the minimum initial horizontal velocity v that the ball must have to get over the net.

[1 mark]

$v =$ _____ m s^{-1}



0 6 . 3

Later in the game, the ball has a horizontal component of velocity of 42 m s^{-1} and a downward vertical component of velocity of 26 m s^{-1}

Determine the resultant velocity of the ball.
You may use a calculation or a scale drawing.

[4 marks]

magnitude of resultant velocity = _____ m s^{-1}

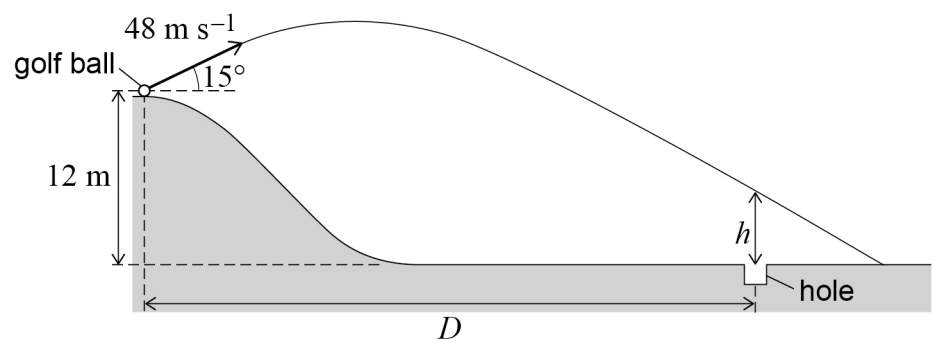
direction of resultant velocity = _____



Figure 4 shows a golf ball hit from a mound of height 12 m.

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Figure 4



The initial velocity of the ball is 48 m s^{-1} at 15° to the horizontal.
 The golf ball takes 3.0 s to reach a position at vertical height h above the hole.
 D is the horizontal distance from the point where the ball is hit to the hole.
 Ignore the effects of air resistance for this question.

0	8	.	1
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Calculate D .

[2 marks]

$D =$ _____ m



0 8 . 2 Show that the initial vertical component of the ball is approximately 12 m s^{-1} .

[1 mark]

0 8 . 3 Calculate h .

[3 marks]

$h =$ _____ m

6

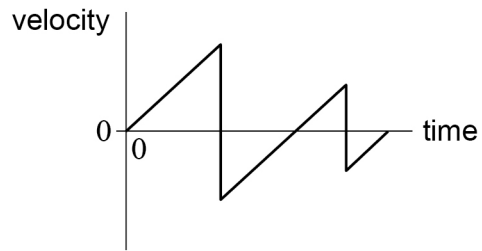
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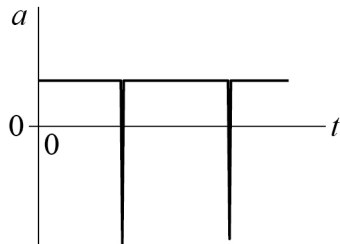
The graph shows the variation with time of the velocity of a bouncing ball.
The downward direction has been chosen to be positive.



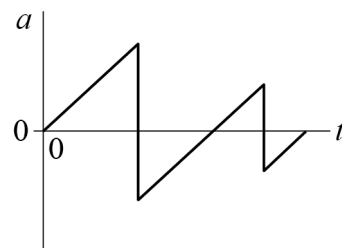
Which graph shows the variation of acceleration a with time t for the ball?

[1 mark]

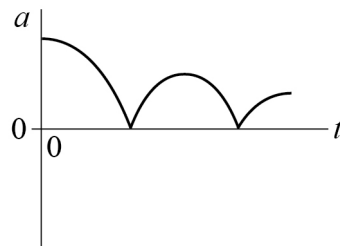
A



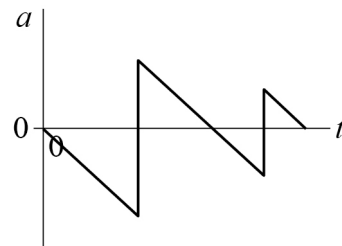
B



C



D



A

☐

B

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C

☐

D

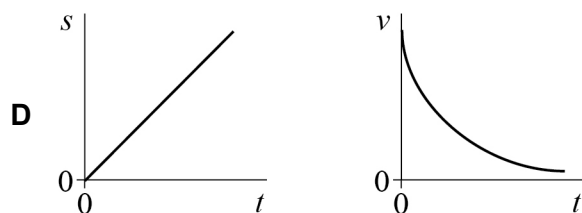
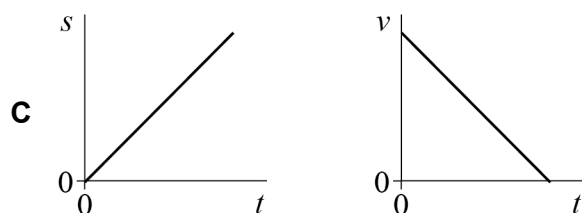
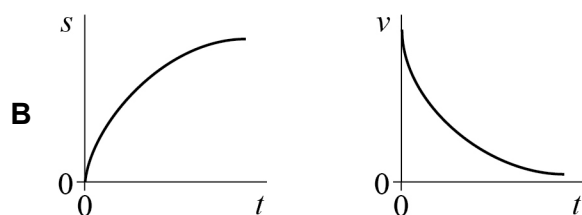
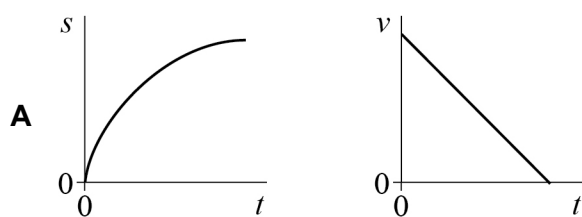
☐


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Brakes apply a constant force to a moving car starting from time $t = 0$

Which pair of graphs shows the variation with time t of displacement s and velocity v ?

[1 mark]



A ☐

B ☐

C ☐

D ☐

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1

8

When a ball is thrown into the air at an angle to the vertical

[1 mark]

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A the forces on the ball are balanced when it is at its highest point.

☐

B the vertical velocity of the ball is zero when it is at its highest point.

☐

C the ball's acceleration increases when it starts to descend.

☐

D the ball's horizontal velocity is maximum when it is at its highest point.

☐

0

1

An object is travelling at constant speed.

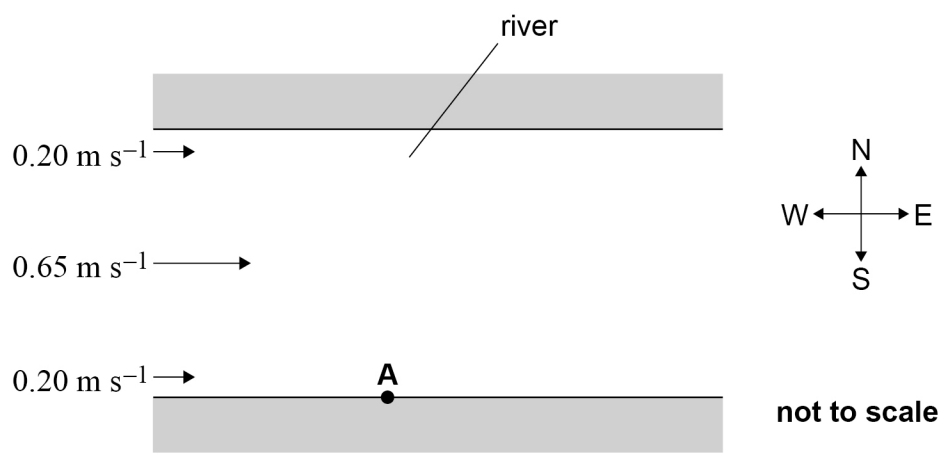
Explain how it can also be accelerating.

[3 marks]

0 5

A river flows from west to east. The velocity of the current in the river varies from 0.20 m s^{-1} near the banks to a maximum of 0.65 m s^{-1} in the middle of the river, as shown in **Figure 1**. The current is always parallel to the bank.

Figure 1



A girl swims from **A** and aims due north. Her speed relative to the water is a constant 1.1 m s^{-1} .

0 5 . 1

Show that the magnitude of the girl's maximum resultant velocity is approximately 1.3 m s^{-1} as she swims across the river.

[2 marks]

0 5 . 2

On different occasions, the girl swims in the river with a speed of 0.90 m s^{-1} relative to the water and with a variety of directions.

State the magnitude and direction of the minimum possible resultant velocity that the girl can have.

[1 mark]

magnitude = _____ m s^{-1}

direction = _____



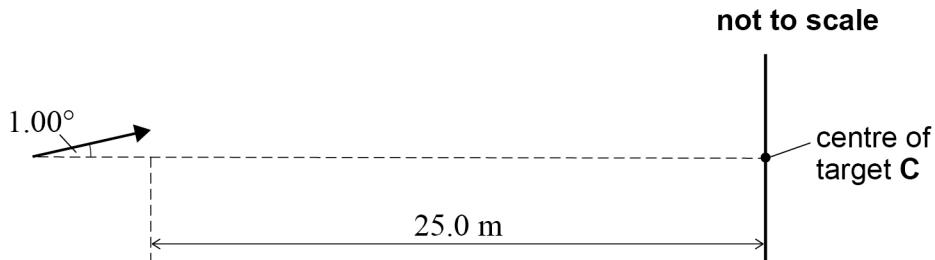
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0 8

An archer fires an arrow at an angle of 1.00° above the horizontal towards a target. The target is 25.0 m away from the point of the arrow when it is fired, as shown in **Figure 3**.

The arrow hits the target 0.260 s after it is fired.
Lift force and drag force are negligible during the flight of the arrow.

Figure 3



0 8

1

Show that the initial vertical component of the velocity of the arrow is approximately 1.7 m s^{-1} .

[2 marks]

0 8

2

The centre **C** of the target is at the same height as the point from which the arrow is fired. The arrow hits the target at a point **P**.

Determine the **vertical** displacement of **P** relative to **C**.

[3 marks]

displacement of **P** relative to **C** = _____ m _____

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In training, an athlete runs from a starting line to a wall and back towards the starting line as shown in **Figure 4**. He starts to run at time $t = 0$

Figure 4

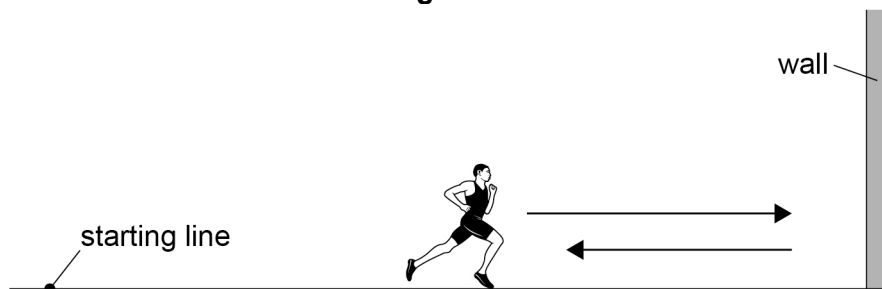
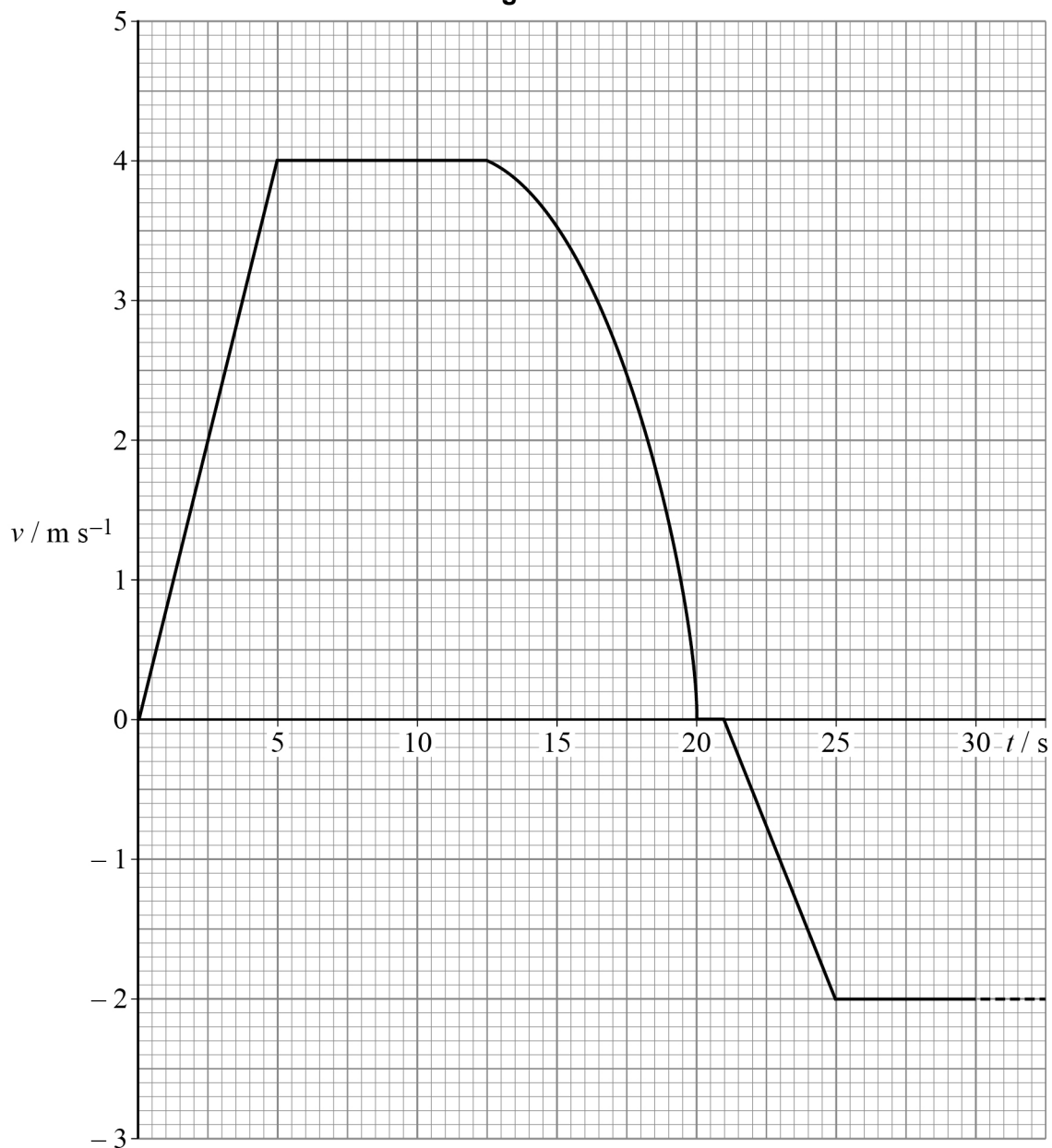


Figure 5 shows how the athlete's velocity v varies with time for the first 30 s of the run.

Figure 5



0 9 . 1 Determine the acceleration of the athlete at $t = 16$ s.

[3 marks]

acceleration = _____ m s^{-2}

0 9 . 2 The athlete's displacement from the starting line is d .

Calculate the magnitude of d at $t = 20$ s.

[3 marks]

$d =$ _____ m

Question 9 continues on the next page

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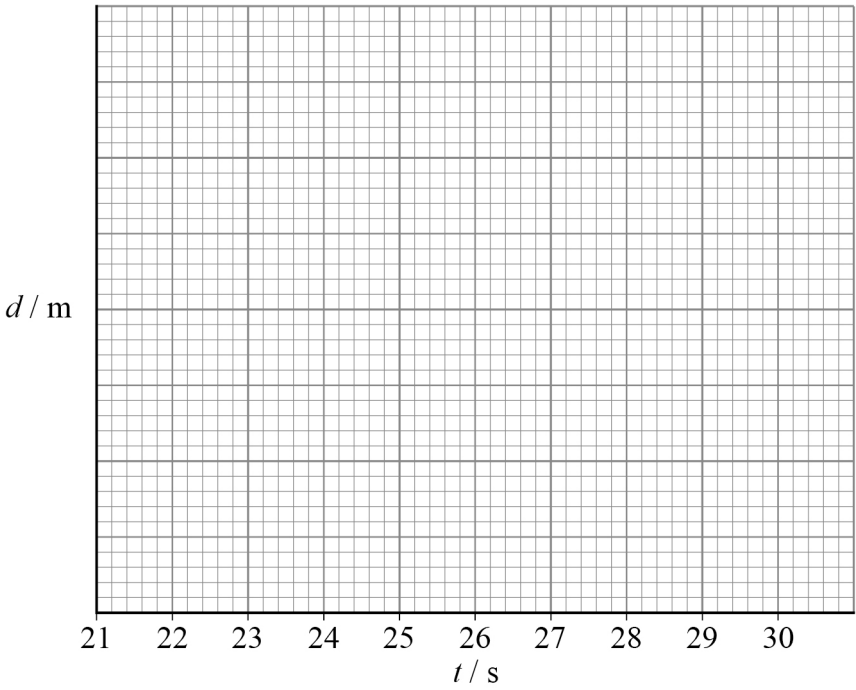


0 9 . 3

Sketch, on **Figure 6**, a graph to show the variation of d with t between $t = 21$ s and $t = 30$ s.

[3 marks]

Figure 6



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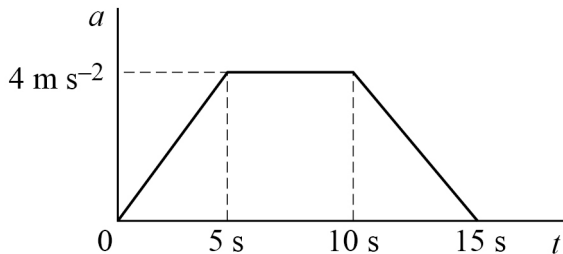
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The graph shows the variation of acceleration a with time t for an object.



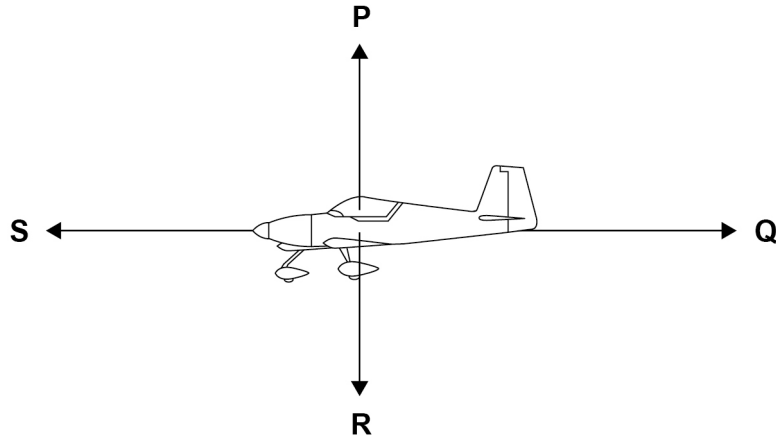
What is the change in velocity of the object due to this acceleration?

[1 mark]**A** 0☐**B** 30 m s^{-1} ☐**C** 40 m s^{-1} ☐**D** 60 m s^{-1} ☐

08

Figure 5 shows an aircraft that is initially in level flight at a constant velocity.

Figure 5



08.1

Identify the forces **P**, **Q**, **R** and **S** acting on the aircraft.

[2 marks]

P _____

Q _____

R _____

S _____

08.2

Force **S** is increased to a new constant value.

State and explain any change in the horizontal and vertical motion of the aircraft due to the increase in **S**.

[3 marks]



On another occasion, the aircraft is flying horizontally when force **P** is increased to 9.87 kN.

The aircraft climbs with constant vertical acceleration.

The weight of the aircraft is 9.60 kN.

0 8 . 3 Show that the vertical acceleration of the aircraft is approximately 0.28 m s^{-2} .

[4 marks]

0 8 . 4 The aircraft climbs with this constant acceleration through a vertical distance of 500 m.

Show that the magnitude of the vertical component of the velocity of the aircraft at the end of the climb is approximately 17 m s^{-1} .

[2 marks]

Question 8 continues on the next page

Turn over ►



0 8 . 5

During the climb, the aircraft has a constant horizontal component of velocity of 70 m s^{-1} .

Calculate the aircraft's resultant velocity at the end of the climb.

[4 marks]

magnitude of resultant velocity = _____ m s^{-1}

angle of the resultant velocity to the horizontal = _____ $^{\circ}$

15

END OF SECTION A



Turn over for Section B

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Section B

Answer **all** questions in this section.

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6

Which is correct?

[1 mark]

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A The area under a distance–time graph is equivalent to speed.

☐

B The area under a speed–time graph is equivalent to acceleration.

☐

C The gradient of a speed–time graph is equivalent to distance.

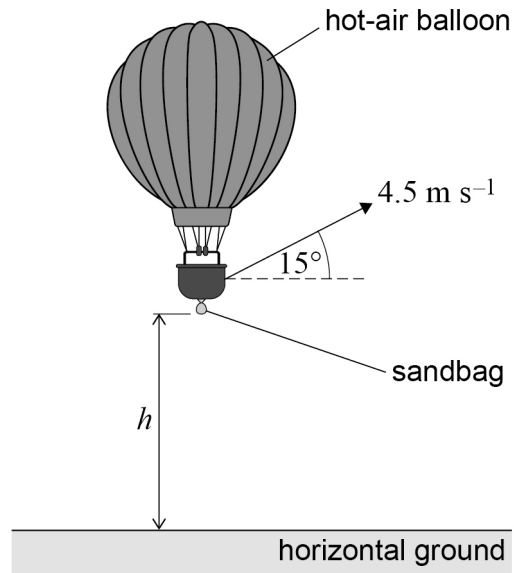
☐

D The gradient of a distance–time graph is equivalent to speed.

☐

Figure 1 shows a hot-air balloon rising at an angle of 15° to the horizontal.

Figure 1



The balloon is travelling at a constant velocity of 4.5 m s^{-1} .
A sandbag is released from the balloon at a height h above the ground.
The sandbag takes 2.5 s to reach the ground.

The air resistance on the sandbag is negligible.

Calculate h .

[4 marks]

$h =$ _____ m



0 3 . 2

The balloon continues to travel at the same constant velocity.

Explain why the sandbag is always vertically below the balloon until the sandbag lands.

[2 marks]

6



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0 6

Figure 5 shows a boy swimming in a pool.
The boy swims in a straight line from point **A** to point **B** and then returns.
The distance from **A** to **B** is 25 m.

Figure 5

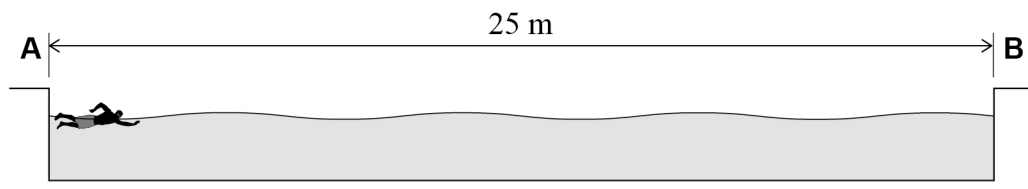
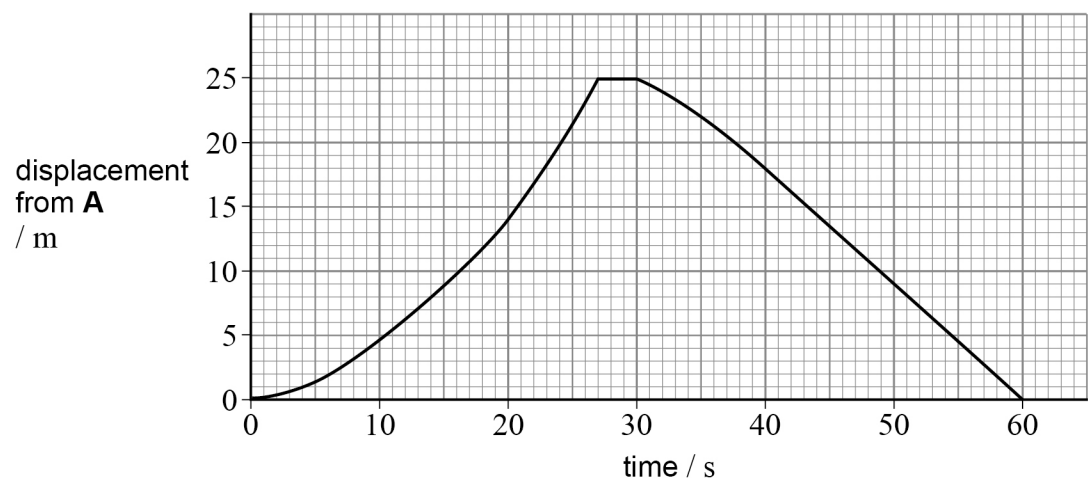


Figure 6 shows the variation with time of the boy's displacement from **A**.

Figure 6



0 6 . 1

Explain how you would use **Figure 6** to determine the maximum speed of the boy.

[2 marks]



0 6 . 2

Calculate the average speed of the boy during the first 50 s of his swim.

[3 marks]

average speed = _____ m s^{-1}

0 6 . 3

Calculate the average velocity of the boy during the first 50 s of his swim.

[2 marks]

magnitude of average velocity = _____ m s^{-1}

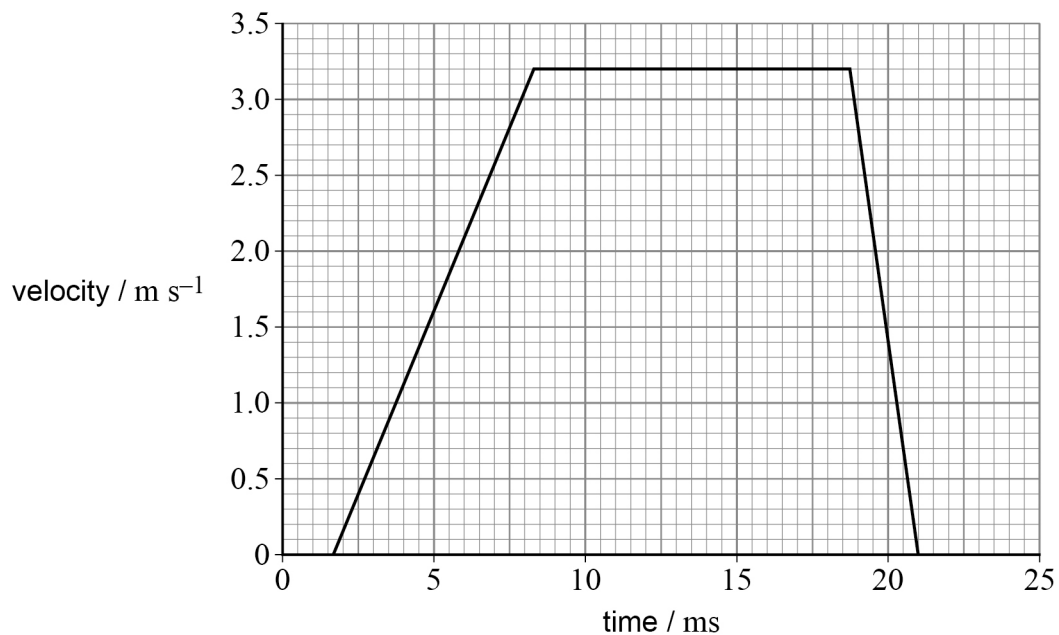
direction of average velocity = _____



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0 4 **Figure 3** is a velocity–time graph for an object travelling in a straight line.

Figure 3



0 4 . 1 Determine the average speed of the object during the time it is moving.

[3 marks]

speed = _____ m s⁻¹

0 4 . 2 The mass of the object is 0.54 kg.

Determine the magnitude of the resultant force required to bring the object to rest.

[2 marks]

magnitude of resultant force = _____ N

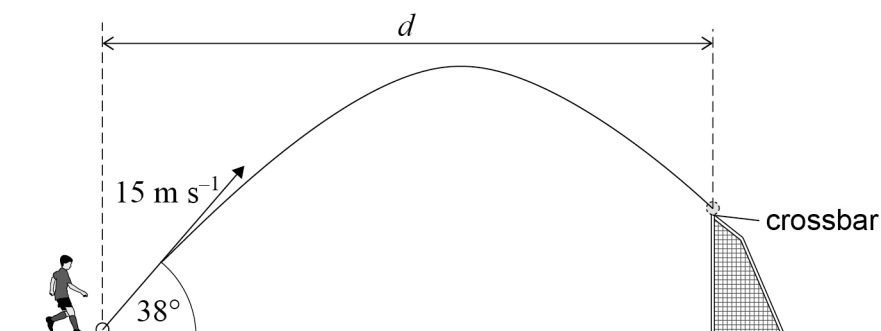
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08

Figure 10 shows a football player kicking a football. The initial velocity of the ball is 15 m s^{-1} at an angle of 38° to the horizontal.

The bottom of the ball hits the top of the crossbar.
Assume that air resistance is negligible.

Figure 10



08.1

Calculate the vertical component of the initial velocity of the ball.

[1 mark]

vertical component of initial velocity = _____ m s^{-1}

08.2

Show that the maximum height gained by the ball is approximately 4.3 m.

[2 marks]

maximum height = _____ m



0 8 . 3

Show that the time between the ball being kicked and it reaching its maximum height is approximately 0.94 s.

[2 marks]

0 8 . 4

The top of the crossbar is at a height of 2.56 m.

Calculate the time between the ball being kicked and it hitting the crossbar.

[3 marks]

time = _____ s

Question 8 continues on the next page

Turn over ►



0 8 . 5

The ball travels a horizontal distance d before hitting the crossbar.

Calculate d .

[2 marks]

$d =$ _____ m

0 8 . 6

To calculate d , it was assumed that air resistance was negligible.

Describe the effect that air resistance will have on the flight of the ball.

[2 marks]

END OF SECTION A



Section B

Answer **all** questions in this section.

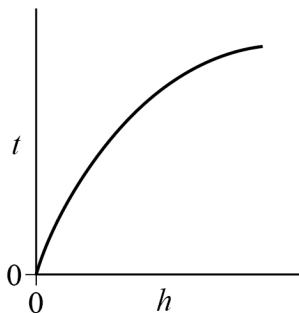
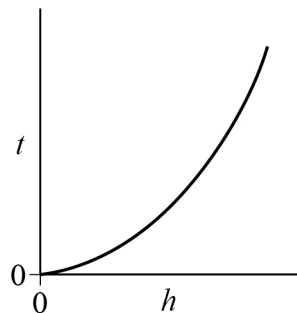
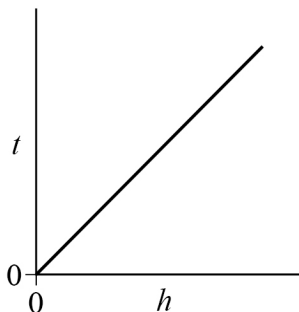
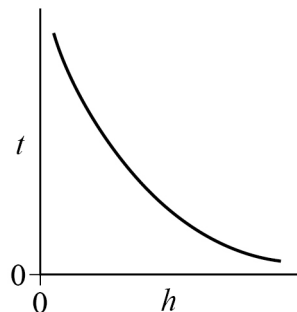
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2 3

An object is released from rest from height h above the ground.
The object takes a time t to reach the ground.

Which graph shows the variation of t with h ?

[1 mark]

A**B****C****D****A****B****C****D**

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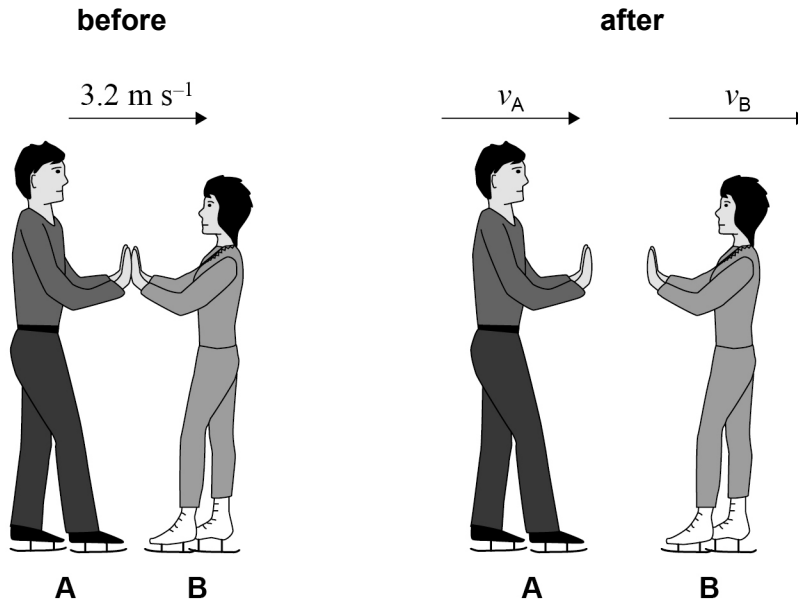
Figure 6 shows two ice skaters **A** and **B**. Both skaters are initially moving horizontally in the same direction at a constant speed of 3.2 m s^{-1} .

A pushes against **B** and applies a constant force of 410 N for 0.15 s .

A and **B** continue to move in the same direction.

The speed of **A** changes to v_A and the speed of **B** changes to v_B .

Figure 6



B has a mass of 56 kg .

Calculate the distance travelled by **B** while the force of 410 N acts.
Assume that no other forces act on **B**.

[3 marks]

distance = _____ m



0 7 . 2 The mass of **A** is greater than the mass of **B**.

Deduce, without calculation, whether **A** or **B** has the greater magnitude of acceleration.

[2 marks]

5

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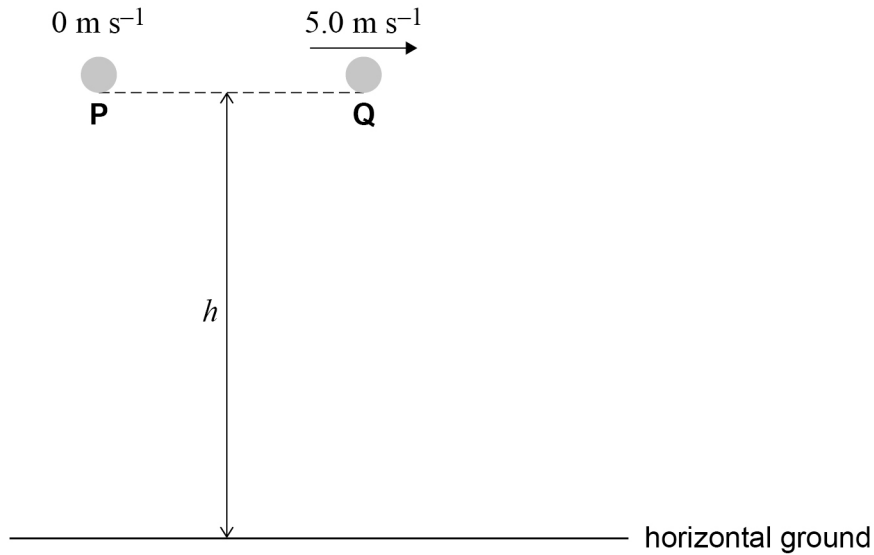
Two identical steel balls **P** and **Q** are released from the same height h above horizontal ground as shown in **Figure 7**.

P is released from rest.

Q is projected with an initial horizontal velocity of 5.0 m s^{-1} .

The air resistance on the steel balls is negligible.

Figure 7



Explain why **P** and **Q** hit the ground at the same time.

[1 mark]



P hits the ground with a speed of 4.2 m s^{-1} .

0 8 . 2

Calculate h .

[1 mark]

$h =$ _____ m

0 8 . 3

Q hits the ground with a speed v and at an angle θ to the vertical.

Determine, using a scale diagram, v and θ .

[4 marks]

$v =$ _____ m s^{-1} $\theta =$ _____ $^{\circ}$

Question 8 continues on the next page

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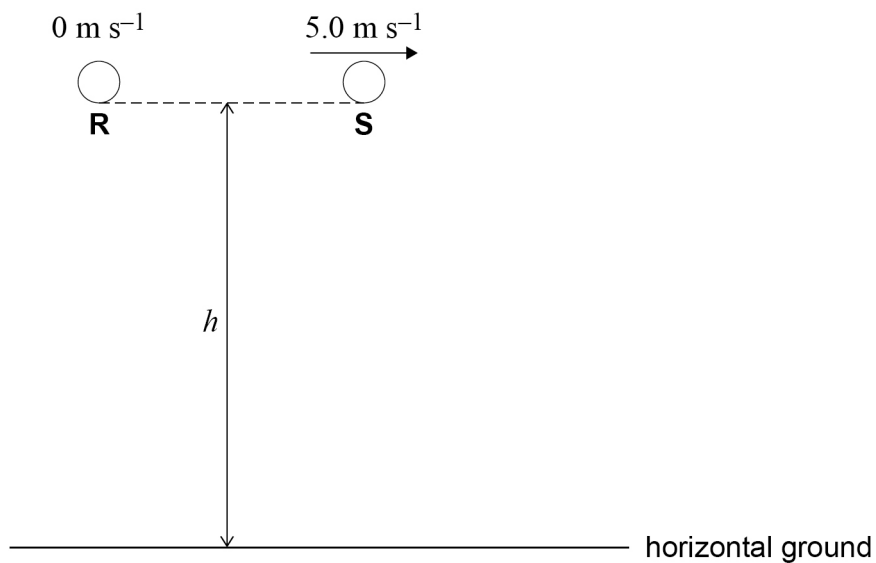


The procedure is repeated using plastic balls **R** and **S** as shown in **Figure 8**.

R and **S**:

- are identical to each other
- have a greater diameter than **P** and **Q**
- have a smaller mass than **P** and **Q**.

Figure 8



0 8 . 4

Explain any difference between the time taken for **steel** ball **P** and **plastic** ball **R** to reach the ground.

[2 marks]



0 8 . 5

The horizontal component of velocity of **steel** ball **Q** when it hits the ground is v_Q .
The horizontal component of velocity of **plastic** ball **S** when it hits the ground is v_S .

Explain the difference between v_Q and v_S .

[2 marks]

10

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1**8**

An electron is accelerated from rest by a constant force F over a distance of 8.5 cm. The final velocity of the electron is $6.2 \times 10^6 \text{ m s}^{-1}$.

What is the magnitude of F ?

[1 mark]

A $2.1 \times 10^{-18} \text{ N}$

☐

B $4.1 \times 10^{-18} \text{ N}$

☐

C $2.1 \times 10^{-16} \text{ N}$

☐

D $4.1 \times 10^{-16} \text{ N}$

☐

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A student does an experiment to determine a value for g , the acceleration due to gravity. A ball is dropped from rest.

The time t taken for the ball to fall a vertical distance s is measured for a range of values of s . The student uses sensors connected to a data logger to measure t .

Table 2 shows the data collected by the student.

Table 2

s / m	t / s
0.300	0.246
0.500	0.322
0.700	0.381
0.900	0.432

09.1

Explain **one** advantage of using a data logger rather than a stopwatch to measure t .
[2 marks]

09.2

Suggest **two** ways in which the student's data could be improved.
[2 marks]

1

2



Table 3 shows the data collected by the student and the student's calculated values for t^2 .

Table 3

s / m	t / s	t^2 / s^2
0.300	0.246	0.06
0.500	0.322	0.10
0.700	0.381	0.15
0.900	0.432	0.19

0 9 . 3

Describe how the recording of t^2 can be improved.

[1 mark]

0 9 . 4

Explain a graphical method to determine g using the data in **Table 3**.

[3 marks]



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A ball is thrown vertically upwards from point **P**. It is caught at **P** as it descends.
The ball is in the air for time t .

What quantity is zero at $\frac{t}{2}$?

[1 mark]

A acceleration

☐

B displacement

☐

C distance travelled

☐

D velocity

☐

2

2

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A person stands on a weighing scale in a lift. When the lift is stationary, the scale reads W .

Which movement of the lift will cause the reading on the scale to be less than W ?

[1 mark]

A downwards at a constant speed

☐

B downwards with an increasing speed

☐

C upwards at a constant speed

☐

D upwards with an increasing speed

☐

0 6

A ball is thrown vertically upwards from an initial height of 1.5 m above horizontal ground. Assume that air resistance is negligible.

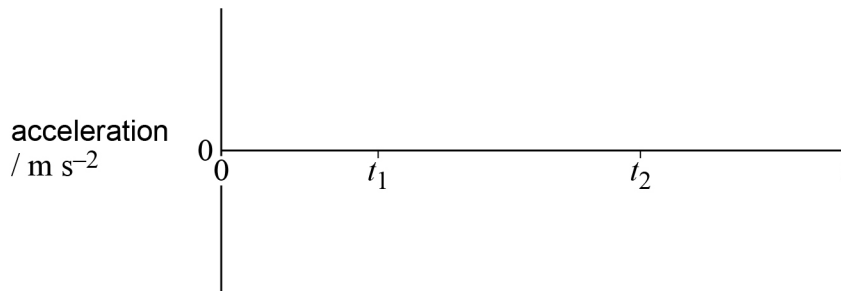
0 6 . 1

The ball is thrown at time $t = 0$. It reaches the top of its path when $t = t_1$. It hits the ground at $t = t_2$ and bounces.

Sketch on **Figure 2** an acceleration–time graph for the ball. You should provide the initial value of the acceleration on the acceleration axis.

[3 marks]

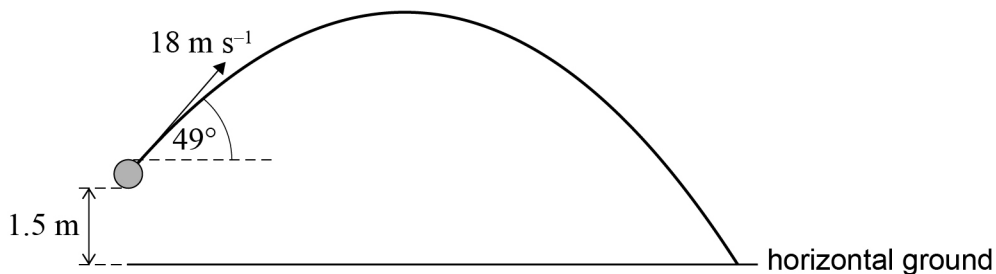
Figure 2



0 6 . 2

The ball is now thrown from the same initial height with a velocity of 18 m s^{-1} at an angle of 49° to the horizontal. **Figure 3** shows the path of the ball.

Figure 3



Calculate the maximum height of the ball above the ground.

[3 marks]

maximum height = _____ m



At one point in its motion, the ball has a minimum speed $v_{\min}$.

0 6 . 3

State and explain the position of the ball when its speed is $v_{\min}$.

[2 marks]

0 6 . 4

Determine $v_{\min}$.

[1 mark]

$v_{\min} =$ _____ m s^{-1}

9

Turn over for the next question

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